VÕ NGỌC MẾN

IOT & EMBEDDED SYSTEMS INTERN

Computer Engineering Technology | HCM-UTE



CONTACT

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Location: Thu Duc City, Ho Chi Minh City

Date of Birth: 27 Feb 2005

EDUCATION

Ho Chi Minh City University of Technology and Engineering (HCM-UTE)

Bachelor's Program - Computer Engineering Technology

2023 - Expected 2027

Relevant coursework:

Internet of Things: 8.0/10

Hardware/Software Co-Design: 8.6/10

Artificial Intelligence: 8.5/10

TECHNICAL SKILLS

Embedded & IoT: ESP32, BLE GATT, UART/I2C/SPI, Wi-Fi, MQTT, NTP, RC522, LCD, sensors, servo

Programming: C/C++, Verilog; hands-on code understanding in BLE Gateway-Sensor and AES-128

Firmware & Debugging: NimBLE-Arduino, FreeRTOS Queue, Arduino Framework, PlatformIO, Serial Debugging

HW/SW & Tools: Zynq-7000, ARM Cortex-A9, AXI4-Lite, Vivado, Vitis, Vivado HLS, KiCad, Git/GitHub, Linux

STRENGTHS

- Sustained focus during coding, testing, and video editing.
- Patience with debugging and iterative testing.
- Responsibility and self-learning.
- Team coordination and collaboration.

ABOUT ME

Computer Engineering Technology student seeking an IoT and Embedded Systems internship. Hands-on understanding of ESP32 BLE Gateway-Sensor firmware and AES-128 HW/SW co-design on Zynq. In other academic projects, I defined requirements, coordinated AI coding agents, tested builds, reproduced defects, and validated end-to-end behavior.

FEATURED PROJECTS

ESP32 BLE Wireless Sensor Network with Failover

Hands-on Implementation & Testing | Gateway-Sensor Focus

Tech: ESP32 DevKit V1, C/C++, Arduino Framework, PlatformIO, NimBLE-Arduino, BLE GATT, FreeRTOS Queue, Serial Debugging

- Implemented and tested Gateway GATT server and Sensor client with service UUID discovery, fixed 10-byte packets, and Write With Response ACKs.
- Queued BLE callback data for loop processing; analyzed sequence numbers, packet loss, duplicates, PDR, RSSI, and Serial CSV logs.
- Validated direct transmission and failover on 3 ESP32 boards: switched after 3 missed ACKs and retried Gateway after 10 successful Relay transmissions. Tested blocked connections and asynchronous disconnects.

https://github.com/menvo-embedded/esp32-ble-wsn-failover

Warehouse Access Control Using RFID & Computer Vision

Project Lead | AI-Assisted Implementation & System Validation | 2-member Project

Tech: ESP32, C/C++, RC522, MQTT, LCD, LED, buzzer, servo, Wi-Fi, NTP, Python, OpenCV, YOLO, Re-ID, InsightFace, Flask, SQLite, KiCad

- Defined requirements, access workflows, test scenarios, and acceptance criteria; applied understanding of AI model training and evaluation workflows.
- AI coding agents generated most or all application code. Assembled hardware, ran builds, tested hardware/software behavior, and reproduced defects for agent fixes.
- Validated end-to-end operation and abnormal access scenarios through repeated testing.

Pipelined AES-128 Hardware Accelerator on Zynq SoC

Team Lead & Primary Technical Contributor | 5-member Project

Tech: Verilog, Vivado, Zynq-7000, ARM Cortex-A9, AXI4-Lite, C, Vivado HLS

- Designed iterative and 10-stage pipelined AES-128 cores; integrated AXI4-Lite with ARM Cortex-A9 and developed a C software-side driver.
- Verified FIPS-197 test vectors and timing closure; achieved 12.8 Gbps steady-state core throughput at 100 MHz in RTL simulation.
- Compared implementation trade-offs between RTL and Vivado HLS.

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WORK EXPERIENCE

Part-Time Video Editor

Accounting Services Company

Approximately 2 months in 2026

Produced approximately 15 short videos for the company's Facebook channel with AI assistance.

Personally reviewed pacing, subtitles, audio, visuals, and final exports.

Service Crew | Highlands Coffee

11/2025 - 04/2026

Trained and worked across sales/order taking, barista, and customer service roles.

Followed store procedures; supported shift handovers and team coordination.

ENGLISH

Basic communication. I can understand familiar questions when spoken slowly and respond using simple English. I can read basic English documents and am currently improving my technical vocabulary and grammar.

AI-ASSISTED DEVELOPMENT

Requirements breakdown, prompt design, agent coordination, iterative testing, bug reproduction, debugging feedback, and output validation.

ADDITIONAL SKILLS

Application validation: Python, OpenCV, YOLO, Re-ID, InsightFace, Flask, SQLite

Mobile & Web: Flutter/Dart, BLoC, Supabase Auth, PostgreSQL, Storage, HTML/CSS/JavaScript

Integration testing: C++17, C# .NET 8, P/Invoke, REST/JSON, CMake

Workflow: AI model training and evaluation workflow; hardware/software integration; video quality review

AI-ASSISTED ACADEMIC PROJECTS

For the projects below, I defined requirements and test cases, directed AI coding agents, ran builds and tests, reproduced defects, and validated end-to-end behavior. AI generated most of the application code.

LuxeStay Hotel Chain Management System

Academic Course Project | 2026

Tech: C++17, C# .NET 8, P/Invoke, REST/JSON, HTML/CSS/JavaScript, CMake

- Defined integration requirements for the C++ business core, C# API gateway, and web front end; directed AI coding agents to deliver the application.
- Ran builds and tested P/Invoke, REST/JSON, and business-state workflows; reproduced defects and validated fixes.

IoT Health Monitoring & Preliminary AI Prediction

Academic Course Project | 2026

Tech: ESP32, C/C++, Heart Rate/SpO2 Sensors, MQTT/HTTP, Python, Machine Learning, Web Dashboard

- Defined sensor, data-transmission, dashboard, and preliminary ML workflows; coordinated AI coding agents to generate code.
- Ran system tests and checked sensor data, transmission behavior, dashboard output, and preliminary predictions.

Forest Field Data & Carbon Management Platform

Academic Course Project | 2026

Tech: Flutter, Dart, BLoC, Supabase Auth, PostgreSQL, Storage, HTML/CSS/JavaScript

- Defined mobile/web requirements and coordinated the Firebase-to-Supabase migration using AI coding agents.
- Tested authentication, GPS check-ins, field logbooks, photo evidence, and role-based workflows; reproduced defects and validated fixes.

HW/SW Co-Design on Zynq SoC

Academic Course Project | 2026

Tech: ZedBoard, Zynq-7000, Vivado, Vitis, C/C++, Verilog, AXI GPIO, HLS, HDL Coder

- Coordinated AI-assisted coursework involving AXI GPIO LED control and custom IP.
- Ran simulation, synthesis, hardware export, and software-controlled IP verification; checked outputs and validated behavior.